

Machine Learning Test

Q1. Predict company profit from advertising & operational spend.

 **Dataset:** 50_Startups.csv (columns: R&D Spend, Administration, Marketing Spend, State, Profit)

Write a Python program to:

Q2. Analyse how TV advertising budget affects sales revenue.

 **Dataset:** advertising 8.csv (columns: TV_Advertising_Budget, Sales)

Write a Python program to:

Q3. Predict car selling price using multiple features.

 **Dataset:** car.csv (columns: Car_Name, Year, Selling_Price, Present_Price, Kms_Driven, Fuel_Type, Seller_Type, Transmission, Owner)

Write a Python program to:

Q4. Predict insurance medical charges based on patient data.

 **Dataset:** insurance.csv (columns: age, sex, bmi, children, smoker, region, charges)

Write a Python program to:

Q5. Predict student academic performance index.

 **Dataset:** Student_Performance.csv (columns: Hours Studied, Previous Scores, Extracurricular Activities, Sleep Hours, Sample Question Papers Practiced, Performance Index)

Write a Python program to:

Q6. Predict whether a user will purchase a product based on age & salary.

 **Dataset:** Social_Network_Ads_logistic.csv (columns: User ID, Gender, Age, EstimatedSalary, Purchased)

Write a Python program to:

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Q7. Predict employee attrition (will they leave the company?).

 **Dataset:** Logistic-HR-Employee-Attrition.csv (target column: Attrition — Yes/No)

Write a Python program to:

Q8. Predict Titanic passenger survival.

 **Dataset:** titanic.csv (columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked)

Write a Python program to:

Q9. Predict loan approval using applicant data

 **Dataset:** logistic_loan_approve.csv

Write a Python program to:

Q10. Build a credit card fraud detection classifier.

 **Dataset:** Logisite_Creditcard.csv (large file — sample 10,000 rows using `nrows=10000`)

Write a Python program to:

Q11. Classify whether a user purchased a product using KNN.

 **Dataset:** Social_Network_Ads_logistic.csv (Age, EstimatedSalary → Purchased)

Write a Python program to:

Q12. Predict wine quality using KNN.

 **Dataset:** winequality.csv (columns: fixed acidity, volatile acidity, ..., quality, Id)

Write a Python program to:

Q13. Build a KNN regressor to predict car fuel efficiency

 **Dataset:** auto-mpg.csv (columns: mpg, cylinders, displacement, horsepower, weight, acceleration, year, origin)

Write a Python program to:

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Q14. Predict employee attrition using KNN and compare with Logistic Regression.

 **Dataset: Logistic-HR-Employee-Attrition.csv** (target: Attrition)

Write a Python program to:

Q15. Use KNN to predict student pass/fail based on academic data

 **Dataset: Student_Performance.csv** (target: Performance Index — classify as Pass if ≥ 60 , else Fail)

Write a Python program to:

Q16. Build a Decision Tree to predict Titanic survival

 **Dataset: titanic.csv** (target: Survived)

Write a Python program to:

Q17. Predict whether a customer will buy insurance using a Decision Tree

 **Dataset: insurance.csv** (build binary target: charges > 15000 $\rightarrow$ 1 (High), else $\rightarrow$ 0 (Low))

Write a Python program to:

Q18. Predict diamond price tier using a Decision Tree.

 **Dataset: diamonds.csv** (columns include: carat, cut, color, clarity, depth, table, price, x, y, z)

Write a Python program to:

Q19. Predict wine quality category using a Decision Tree

 **Dataset: winequality.csv** (target: quality — convert to Low/Medium/High)

Write a Python program to:

Q20. Predict employee salary range using a Decision Tree Regressor.

 **Dataset: Logistic-HR-Employee-Attrition.csv** (target: MonthlyIncome)

Write a Python program to:

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Q21. Segment mall customers into groups using K-Means.

 **Dataset:** Mall_Customers 2.csv (columns: CustomerID, Gender, Age, Annual Income (k\$), Spending Score (1-100))

Write a Python program to:

Q22. Segment customers based on full profile using K-Means

 **Dataset:** customer_segmentation_data.csv (columns: age, income, spending_score, membership_years, purchase_frequency, last_purchase_amount, etc.)

Write a Python program to:

Q23. Apply DBSCAN to detect outlier customers in mall data

 **Dataset:** Mall_Customers 2.csv (Annual Income, Spending Score)

Write a Python program to:

Q24. Cluster global air pollution data to find high-risk zones.

 **Dataset:** global_air_pollution_dataset.csv

Write a Python program to:

Q25. Tune DBSCAN parameters and evaluate cluster quality using Silhouette Score

 **Dataset:** customer_segmentation_data.csv (use age, income, spending_score)

Write a Python program to:

Q26. Predict Titanic survival using Random Forest and compare with Decision Tree

 **Dataset:** titanic.csv (target: Survived)

Write a Python program to:

Q27. Predict employee attrition using Random Forest — tune hyperparameters

 **Dataset:** Logistic-HR-Employee-Attrition.csv (target: Attrition)

Write a Python program to:

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Q28. Build a Random Forest Regressor to predict car selling price

 **Dataset: car.csv** (target: Selling_Price)

Write a Python program to:

Q29. Predict insurance charges using Random Forest with cross-validation

 **Dataset: insurance.csv** (target: charges)

Write a Python program to:

Q30. Build a full ML pipeline with Random Forest and feature selection

 **Dataset: winequality.csv** (binary target: quality $\geq 6 \rightarrow$ Good Wine)

Write a Python program to: